

Two Samples, One Difference

Unit 4 · Topic 4.6 · THE PROBLEM

Suppose parcel pickup waits have the shown population parameters, measured in minutes. There are 600 North visits and 1,000 South visits. North waits are approximately normal; South waits are strongly right-skewed. Independent SRSs are taken without replacement, and the service wants $P(\bar{X}_N - \bar{X}_S > 2.5)$.

Rina: “South is skewed, so I would not use a normal model for the difference.”

Oscar: “North is normal and the combined sample size is 60, so the normal model is valid; the probability is 0.0273.”

Locker samples

Locker	n	μ	σ
North	12	6.0	1.8
South	48	4.8	3.0

FIRST TAKE — one sentence before you work the parts

Would you check each place’s sample size separately or add them together? Give one reason from the study.

- DOK 1** (a) State the mean of $\bar{X}_N - \bar{X}_S$ and translate the event above into context.
- DOK 2** (b) Verify both 10 percent conditions, compute the sampling-distribution SD, and calculate the probability under a normal model.
- DOK 3** (c) In 4–5 sentences, decide whether 0.0273 is supported and explain what each student overlooks. Justify each sample mean using its own shape and size. Then consider $n_N = 48$ and $n_S = 8$: explain why their total alone cannot justify the same normal-model claim.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.